

From Stethoscope to Scalpel: Making Feature Geometry Prescriptive for Fine-Tuning

Project page: [SPAR project](#)

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Question 1

Base and instruction-tuned variants of a model show nearly identical SAE feature-divergence-by-layer profiles, yet respond oppositely to late-layer-only LoRA. Propose one hypothesis for why, and one experiment costing at most \$500 in compute that could falsify it. Maximum 300 words.

Although base and instruction-tuned models show similar SAE feature-divergence-by-layer profiles, the same task-relevant representations may participate differently in the computational mechanisms of the two models. My hypothesis is that instruction tuning changes the causal use of these features, which could explain why late-layer LoRA works well in the instruction-tuned model but not in the base model.

To test this, I would first identify task-relevant SAE features in late layers and match corresponding features between the base and instruction-tuned models. For a set of task examples, I would measure baseline task accuracy and then increase the activation of these features in both models. I would measure the resulting change in task accuracy, as well as the change in probability assigned to the correct answer.

The hypothesis predicts that identical feature interventions will produce different performance gains in the base and instruction-tuned models, indicating that a similar SAE feature-divergence-by-layer profile need not imply similar causal use of those features.

The null hypothesis is that matched feature interventions produce comparable effects in the two models. If performance and correct-answer probability change by similar amounts across features, prompts, and late layers, while late-layer LoRA still works only for the instruction-tuned model, this would falsify the hypothesis that different causal readout of these SAE features explains the LoRA discrepancy.

Question 2

The Jacobian lens reads single-token concepts through an averaged Jacobian; SAEs decompose activations into learned features. Name one question about fine-tuning each instrument can answer that the other cannot. Maximum 150 words.

SAEs: SAEs reveal interpretable features in each layer's high-dimensional activation space. After instruction tuning, a common SAE can show which existing features become more or less active and whether their layer-wise locations change. If the SAE generalizes to both models [1], matched features can also be manipulated causally to test whether fine-tuning changes their effect on task performance.

Jacobian lens: The Jacobian lens reveals which single-token concepts at an intermediate layer are most strongly connected to downstream output computation [2]. Comparing base and fine-tuned models can therefore show whether fine-tuning changes which concepts downstream computation uses, and the layer or stage at which task-relevant concepts become accessible.

Question 3

Link to a technical writing sample: a report, blog post, thesis chapter, or anything showing how you communicate results.

DDM notebook, University of Coimbra. This is a Colab notebook that I used when I was a teaching assistant for a one-day workshop at the University of Coimbra. It covers the basics of drift diffusion models and Bayesian fitting, and includes both exercises and prompts that ask readers to pause and reason about the material [3].

References

- [1] Lieberum et al., “Gemma Scope: Open Sparse Autoencoders Everywhere All At Once on Gemma 2,” especially Figure 8. [arXiv](#)
- [2] Lindsey et al., “Verbalizable Representations Form a Global Workspace in Language Models.” [Transformer Circuits](#)
- [3] [DDM teaching notebook, University of Coimbra](#)